# A reproducible computational framework for decoupled wildfire ignition and propagation benchmarking

D2 targets | gated M3/M1 inference | open benchmark against Canadian FWI and INPE

## INPUTS

- **VIIRS active fires**
- Suomi-NPP, NOAA-20/21 375-m detections
- **ERA5-Land weather**
- VPD, rain, wind, FFMC/ISI
- **MapBiomas C2 fuel**
- Pasture, savanna, forest
- **Causal fire history**

## D2 TARGET OPERATOR

**Primary ignition**

- fire at cell i, day t
- no t-1 fire within 25 km
- no same-cell fire in 10 d

**Tactical propagation**

- fire at cell i, day t
- within 25 km of a t-1 fire

*The two target domains are disjoint.*

D2 is applied once and reused for training, test and alerts.

## BIOPHYSICAL GATE

*Applied before inference*

- **fuel > 0**
- **VPD ≥ 0.8 kPa**
- **48-h rain < 10 mm**

Removes water, urban surfaces and saturated fuels from inference.

*The gate is not a label filter.*

## DECOUPLED INFERENCE

**M3 hybrid ignition**

weather + fuel + ecotone
+ anthropogenic priors
no t-1 fire input

**M1 fallback**

weather + fuel only
used where fire history
is absent

**Propagation**

causal t-1 kernel:
distance, counts and FRP

*Decoupled, not persistence.*

## EVALUATION + OUTPUTS

- Temporal test 2024–2025
- 50-km spatial blocks
- Grouped bootstrap
- Lead-time analysis
- Cost-loss and conformal

**Alert maps**
**Benchmark tables**
**Open code and data**

*Every result is tied to a frozen split.*

**Contribution: an auditable target operator, gated inference and evaluation harness — not a new boosting algorithm.**